

LAMPANG, Thailand

WMO: 483280

Lat: 18.2781N

Lon: 99.5064E

Elev: 800

StdP: 14.28

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	55.0	57.7	47.9	50.9	80.3	50.9	57.0	76.1	8.9	80.3	7.6	81.4	0.2	120	0.294

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	24.4	102.6	75.0	100.4	74.6	98.2	74.7	81.2	89.9	80.3	89.8	79.7	89.2	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	156.2	84.7	78.3	151.6	84.0	77.4	146.6	83.2	45.8	90.0	44.8	89.9	44.1	89.8	93.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.4	8.6	7.3	DB	50.1	105.8	3.6	2.0	47.5	107.2	45.3	108.4	43.3	109.5	40.7	111.0	
			WB	49.3	84.4	3.5	3.0	46.7	86.6	44.7	88.3	42.7	90.0	40.1	92.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.2	74.3	78.2	83.4	87.3	85.4	84.4	83.3	82.7	82.5	81.3	78.1	74.0
	DBStd	5.12	3.56	3.47	3.51	3.50	3.66	2.23	2.34	2.02	1.97	2.43	3.44	4.26
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	4	1	0	0	0	0	0	0	0	0	0	0	3
	CDD50	11401	753	790	1035	1119	1098	1031	1032	1013	975	969	843	743
	CDD65	5930	289	370	570	669	633	581	567	548	525	504	393	281
	CDH74	63756	2592	4041	7107	8924	7552	6679	6123	5527	5095	4556	3292	2268
	CDH80	29818	1075	2109	4135	5316	3836	2959	2472	2054	1917	1783	1324	838
(14) Wind	WSAvg	1.3	0.9	1.1	1.4	1.7	1.6	2.0	1.9	1.5	1.1	0.9	0.8	0.8
(15) Precipitation	PrecAvg	44.6	0.2	0.3	0.9	2.5	6.1	5.0	5.4	8.8	8.8	5.0	1.3	0.3
	PrecMax	59.8	2.3	2.2	2.2	9.0	12.2	9.1	13.1	16.3	12.9	10.1	5.3	1.1
	PrecMin	28.7	0.0	0.0	0.0	0.0	0.9	1.6	2.2	5.0	4.5	1.9	0.0	0.0
	PrecStd	8.3	0.6	0.6	0.7	2.3	2.8	2.0	2.9	2.7	2.7	2.1	1.4	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.5	99.0	103.5	106.3	104.0	98.3	96.4	94.8	94.9	94.9	94.4	92.4
		MCWB	69.6	70.7	72.0	75.5	76.9	78.8	78.4	78.6	78.8	78.9	75.5	72.9
	2%	DB	91.1	96.5	100.9	104.0	100.3	95.3	93.6	92.6	92.8	92.6	91.8	89.7
		MCWB	69.8	69.7	71.8	75.1	77.9	78.5	78.3	78.8	79.1	78.2	75.2	72.2
	5%	DB	89.0	94.3	98.9	101.6	97.1	93.3	91.8	90.8	91.0	90.7	89.8	87.6
		MCWB	69.7	69.9	71.6	75.5	78.2	78.6	78.1	78.5	78.7	77.9	75.2	71.8
	10%	DB	86.5	91.7	96.7	98.9	94.5	91.3	89.9	89.0	89.1	88.7	87.6	85.2
		MCWB	69.4	70.0	71.8	75.9	78.1	78.3	77.8	78.2	78.5	77.4	74.5	70.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.6	76.4	79.8	83.9	83.7	82.1	81.2	81.0	81.4	80.7	79.2	77.2
		MCDB	85.8	90.3	89.4	94.4	90.6	89.1	88.8	88.9	89.8	89.8	87.8	87.1
	2%	WB	73.7	74.7	77.0	80.1	80.7	80.4	80.0	80.0	80.3	79.7	78.0	75.3
		MCDB	84.1	87.8	90.9	92.4	91.9	90.0	88.9	88.2	88.8	88.6	86.7	84.9
	5%	WB	72.4	73.5	75.6	78.5	79.6	79.6	79.1	79.2	79.6	79.0	77.0	73.8
		MCDB	82.8	86.5	89.9	93.3	91.1	89.0	88.1	87.3	87.8	87.7	85.6	82.8
	10%	WB	71.1	72.2	74.7	77.5	78.9	78.5	78.5	78.7	79.0	78.3	76.0	72.6
		MCDB	81.7	85.1	88.9	92.1	90.3	88.3	87.1	86.6	86.8	86.3	84.0	81.2
(35) Mean Daily Temperature Range	5% DB	MDBR	27.3	30.0	28.2	24.4	18.1	15.3	14.1	13.9	14.8	16.6	20.6	24.1
		MCDBR	30.3	33.2	32.1	28.0	22.5	18.3	17.0	16.6	17.6	19.3	23.0	26.1
		MCWBR	11.1	10.3	8.9	7.1	6.3	5.3	5.3	5.5	5.8	6.7	8.3	9.9
	5% WB	MCDBR	24.0	26.8	26.0	23.4	18.8	16.0	15.1	14.8	15.9	17.2	19.3	22.1
		MCWBR	9.4	9.6	8.8	8.1	6.6	5.7	5.5	5.5	5.9	6.5	7.5	9.0
(40) Clear-Sky Solar Irradiance	taub		0.448	0.532	0.667	0.653	0.501	0.442	0.431	0.445	0.480	0.527	0.447	0.443
	taud		2.173	1.953	1.657	1.746	2.183	2.379	2.407	2.348	2.233	2.058	2.253	2.212
	Ebn at Noon		263	246	218	223	256	270	273	271	262	244	261	260
	Edh at Noon		45	58	81	75	48	39	38	40	45	52	41	42
(44) All-Sky Solar Radiation	RadAvg		1446	1671	1771	1916	1745	1551	1449	1429	1476	1434	1465	1373
(45)	RadStd		73	66	112	106	149	118	100	89	67	81	95	72

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+2	N/A	N/A
Regional (21 neighbors)	+0.47	N/A	N/A	N/A	N/A	+0.27	N/A	N/A	+218	+218

Nomenclature: See separate page